DATA ENGINEERING

Star vs Snowflake Schema Visual Guide

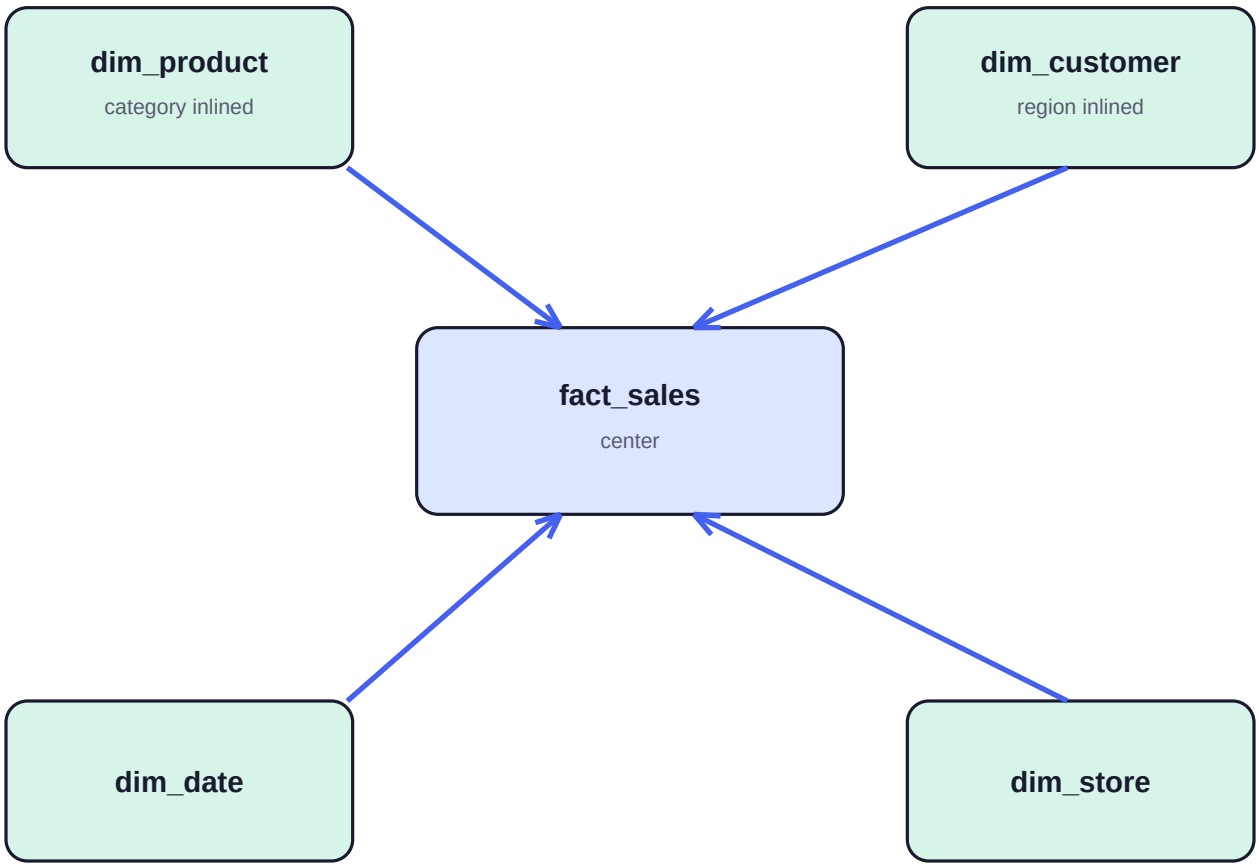
Denormalize or normalize?
The trade-off at the heart of analytics modeling.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Star Schema

Denormalized dimensions, single-hop joins.

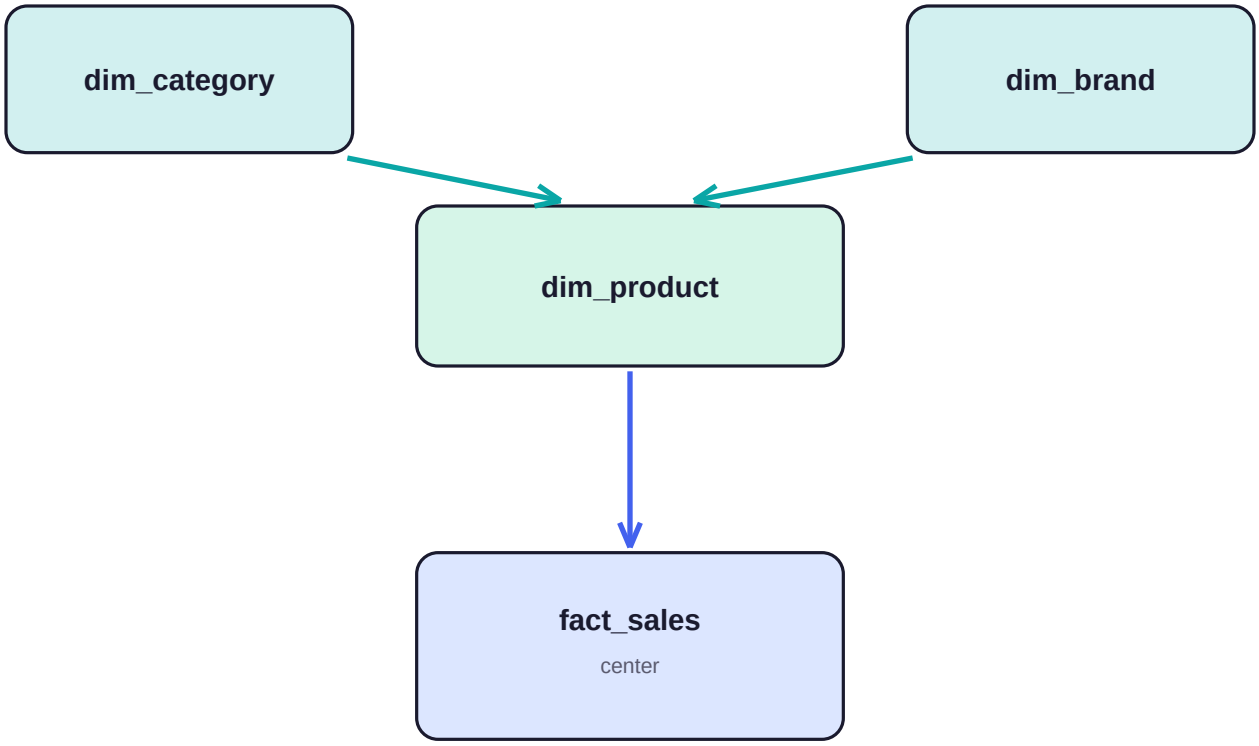


Dimension attributes are flattened into the dimension table — category lives IN dim_product. Redundant storage, trivial queries.

STEP 2/6

The Snowflake Schema

Normalized dimensions, multi-hop joins.



dim_product now points to dim_category and dim_brand. Storage is lean, but 'sales by category' needs two joins — and BI tools must model the chain.

STEP 3/6

The Trade-Off Matrix

What you gain and pay with each shape.

Star wins

- Simple, fast queries
- BI-tool friendly
- Storage is cheap anyway

Snowflake wins

- No update anomalies
- Enforced hierarchies
- Huge shared dimensions

In cloud warehouses storage is the cheapest resource — star's redundancy costs pennies while its simplicity saves engineering hours.

STEP 4/6

Query Pattern Comparison

Same question, different SQL.

Star: one join

```
SELECT p.category, SUM(f.amount)
FROM fact_sales f JOIN dim_product p
  ON f.product_sk = p.product_sk GROUP BY 1;
```

Snowflake: two joins

```
SELECT c.name, SUM(f.amount)
FROM fact_sales f
JOIN dim_product p ON f.product_sk = p.product_sk
JOIN dim_category c ON p.category_sk = c.category_sk
GROUP BY 1;
```

Multiply this by every analyst, every dashboard, every ad-hoc question. Join chains are a tax paid forever.

STEP 5/6

When Snowflaking Makes Sense

Exceptions to the star default.

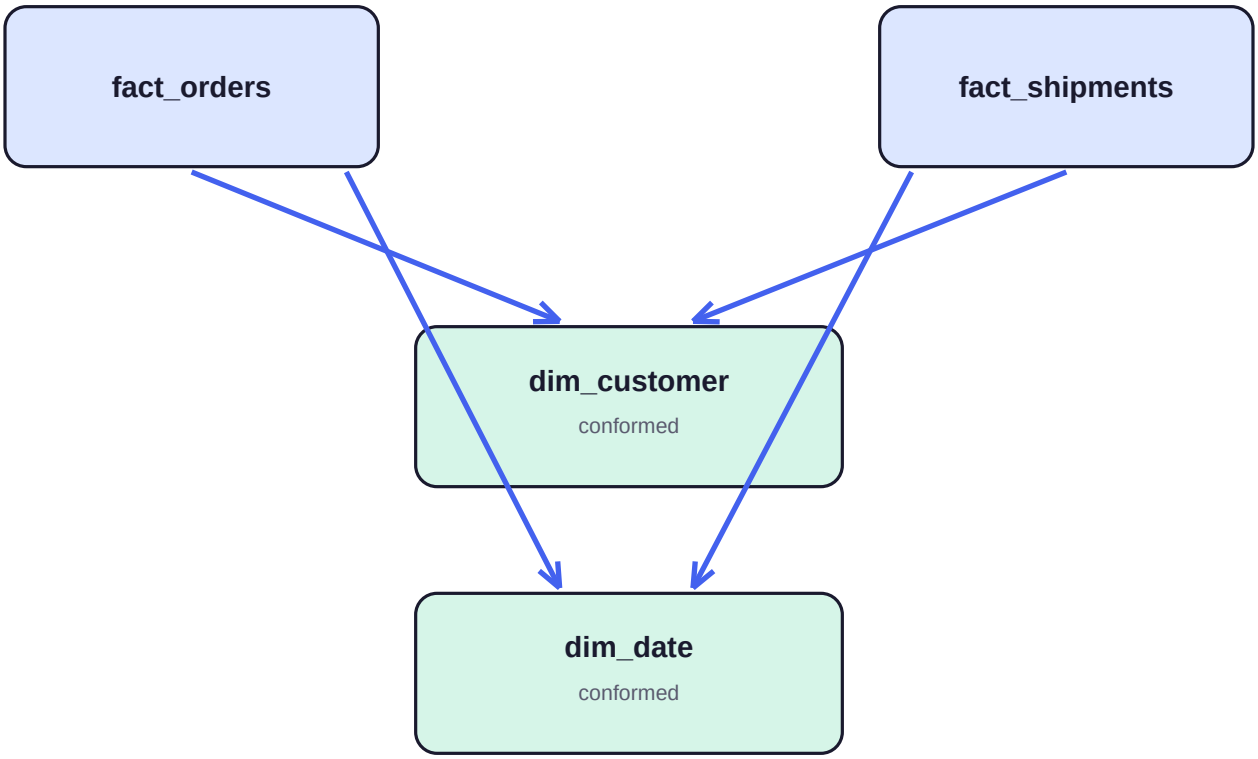
- Massive shared hierarchies (a 10M-row geography tree) where duplication hurts
- Strict write integrity: one place to update a category name
- Source-system fidelity: staging mirrors the OLTP schema before modeling
- Many-to-many bridges (product ↔ promotion) still need bridge tables even in stars

Rule of thumb: model the marts as stars; let staging stay normalized. Snowflake the exception, not the default.

STEP 6/6

Beyond: Galaxy / Constellation

Multiple fact tables sharing conformed dimensions.



Real warehouses are constellations: many fact tables (each a star) sharing conformed dimensions. Drill-across queries work because 'customer' means the same thing everywhere.

STEP ✓/✓

Key Takeaways

Star vs Snowflake Schema — Remember

- Star: denormalized dimensions, single-hop joins — simple queries, BI-friendly, tiny storage cost.
- Snowflake: normalized dimensions, multi-hop joins — lean storage, complex queries.
- In cloud warehouses, storage is cheap: default to star for marts.
- Snowflake the exceptions: giant shared hierarchies, strict write integrity, source-faithful staging.
- Real warehouses are constellations — multiple star facts sharing conformed dimensions.

Keep exploring

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